NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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REFERENCE FILE

TECP 700-700
Materiel Test Procedure 3-2-804*
Aberdeen Proving Ground

27 October 1965

U. S. ARMY TEST AND EVALUATION COMMAND
COMMON ENGINEERING TEST PROCEDURE

IMPRESSIONS AND CASTS OF CANNON BORES

1. OBJECTIVE

The object of this procedure is to instruct personnel in the making of impressions and casts of cannon bores.

2. BACKGROUND

Impressions and casts of cannon bores provide valuable aids for the examination and evaluation of bore conditions. They provide permanent reproduction of defects, erosion conditions and abnormalities in cannon bores, which are an excellent means for making reasonably accurate physical measurements, and are frequently a necessary supplement to visual inspection.

3. REQUIRED EQUIPMENT

a. Heating Device
b. Impression Making Device consisting of:

1. Getta-percha
2. Powdered Graphite
3. Impression Block and Distance Rod of appropriate size
4. Wedges and Wood Rods of appropriate size
5. Face Piece
6. Face Plates as required
7. Sledge
8. Water Container and heated water (minimum of 140°F)

c. Cast Making Device consisting of:

1. Sulfur and Lampblack or
2. Woods Metal
3. Bottle Cork of appropriate size
4. Twisted Double Strand of Wire (approximately 0.05 inch diameter)
5. Container for melted sulfur and/or Woods Metal cast material

d. Steel Dividers and Rule for Measuring
e. Camera and film

For description and/or uses of pertinent required equipment see APPENDIX A.

*Supersedes Ordnance Proof Manual 80-44

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4. REFERENCES

None

5. SCOPE

5.1 SUMMARY

This MTP pamphlet describes methods and techniques for making Gutta-Percha impressions and sulfur or metal-alloy casts of various areas of cannon bores.

5.2 LIMITATIONS

Gutta-percha impressions shall be made in cannons having bore diameters of 20 millimeters or greater. For smaller bores, sulfur or metal-alloy casts shall be used.

6. PROCEDURES

6.1 PREPARATION FOR TEST

Clean and dry the gun bore before an impression or cast is made.

6.2 TEST CONDUCT

6.2.1 Gutta-Percha Impressions

Impressions shall never be made in a heated cannon, whether the heating is caused by firing or by the action of the sun's rays. In warm weather, impression-making shall be scheduled for the early morning hours.

Gutta-percha impressions shall be made by the following procedures:

a. A block of appropriate size, together with its mating wedge shall be selected. The required distance rods and stops shall be assembled to the block.

b. The gutta-percha shall be placed in heated water until it becomes sufficiently pliant, after which it shall be mounted on the impression block, coated lightly with graphite, and placed in the desired location in the gun bore.

c. The wedge shall be driven in to force the gutta-percha to fill the depths of the grooves or defect.

d. After allowing the material to harden (approximately 15 minutes), the wedge and impression shall be removed.

e. Visually inspect the impression. Record observations as to defects, erosion conditions, and other abnormalities existing in the cannon bore.

f. Measure and record defects, depth of erosion, and other abnormalities existing in cannon bore.

g. Photograph impression.

NOTE: Figures 1 and 2 show gutta-percha impressions of conditions of erosion and damage in cannon bore.

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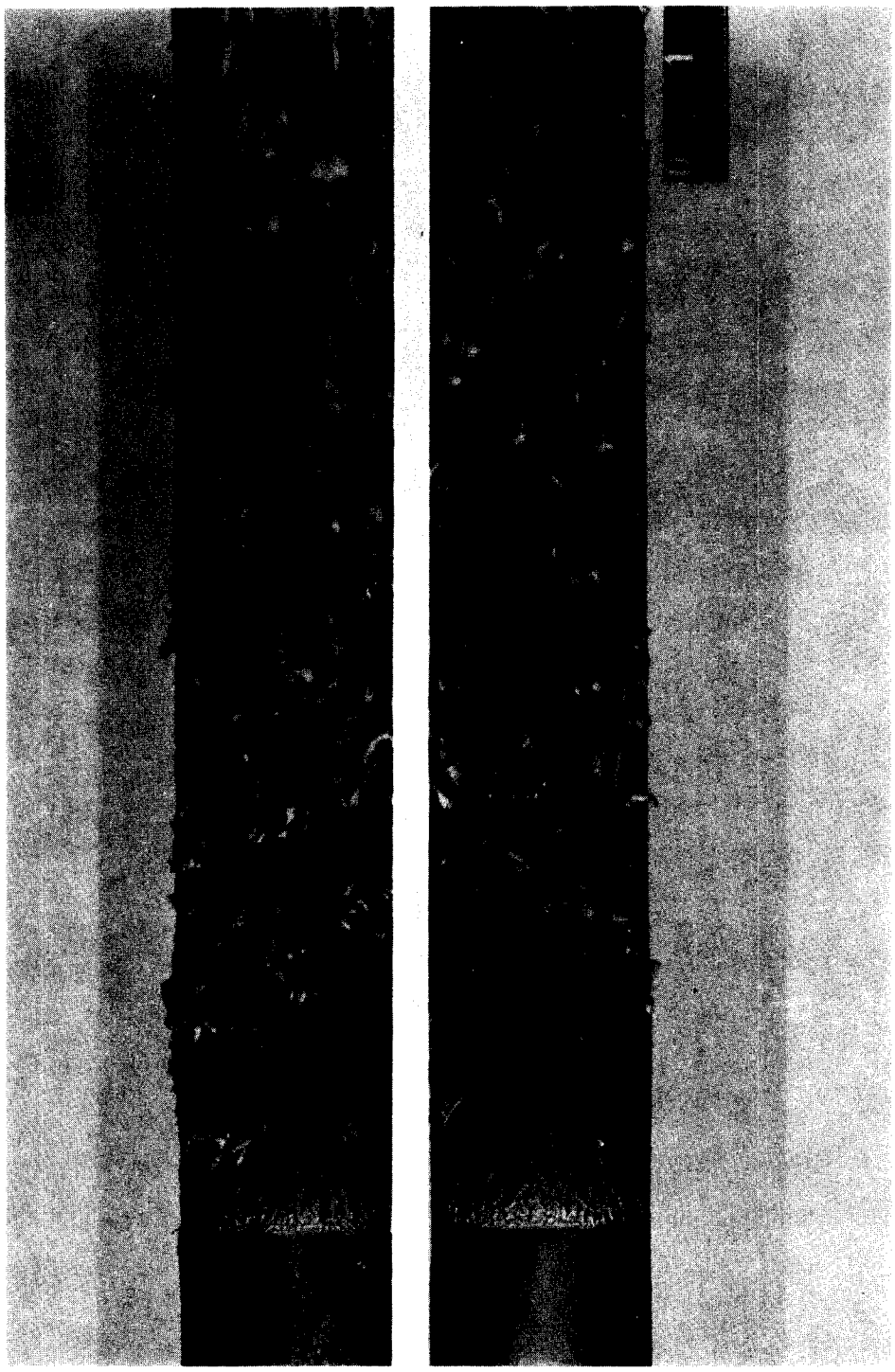


Figure 1. Gutta-Percha Impressions. Impressions Showing Condition of Erosion in Cannon at Origin of Rifling. (Top) - 12 o'clock position in Bore. (Bottom) - 6 o'clock Position in Bore

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Figure 2. Gutta-Percha Impressions. Condition Present in Cannon Tube-105-mm Howitzer. (Top) - Stripped and Raised Lands at Origin of Rifling. (Bottom) - Damage in Bore Caused by Rotating Arming Wires

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6.2.2 Sulfur or Metal Alloy Casts

Sulfur or metal alloy casts shall be made employing the following procedures:

a. For Woods metal only, pre-heat the cannon bore to assure uniform density of casting.

b. For sulfur and Woods metal:

1. Heat the casting material until it liquifies.
2. Insert the twisted wire through the cork, and insert the whole assembly in the gun barrel with the end of the wire extending from the bore.
3. Pour the casting material in the barrel until the desired area is thoroughly filled.
4. Wait for the cast to solidify. Complete solidification should take place within 20 minutes.
5. Visually inspect the cast. Record observations as to defects, erosion conditions, and other abnormalities existing in cannon bore.
6. Measure and record defects, depth of erosion, and other measurable abnormalities existing in cannon bore.
7. Photograph cast.

NOTE: Figure 3 shows casts made of 20 millimeter chambers using sulfur and Woods metal, respectively.

6.3 TEST DATA

6.3.1 Gutta-Percha Impressions

Record visual observations as to defects, erosion conditions, and other abnormalities existing in cannon bore.

Record measurements of defects, depth of erosion and other abnormalities in inches.

6.3.2 Sulfur or Metal Alloy Casts

Record visual observations as to defects, erosion conditions, and other abnormalities existing in cannon bore.

Record measurements of defects, depth of erosion, and other abnormalities in inches.

6.4 DATA REDUCTION AND PRESENTATION

Photographs and recorded measurements of the test weapon casts and impressions shall be retained with the test weapon as part of its permanent log.

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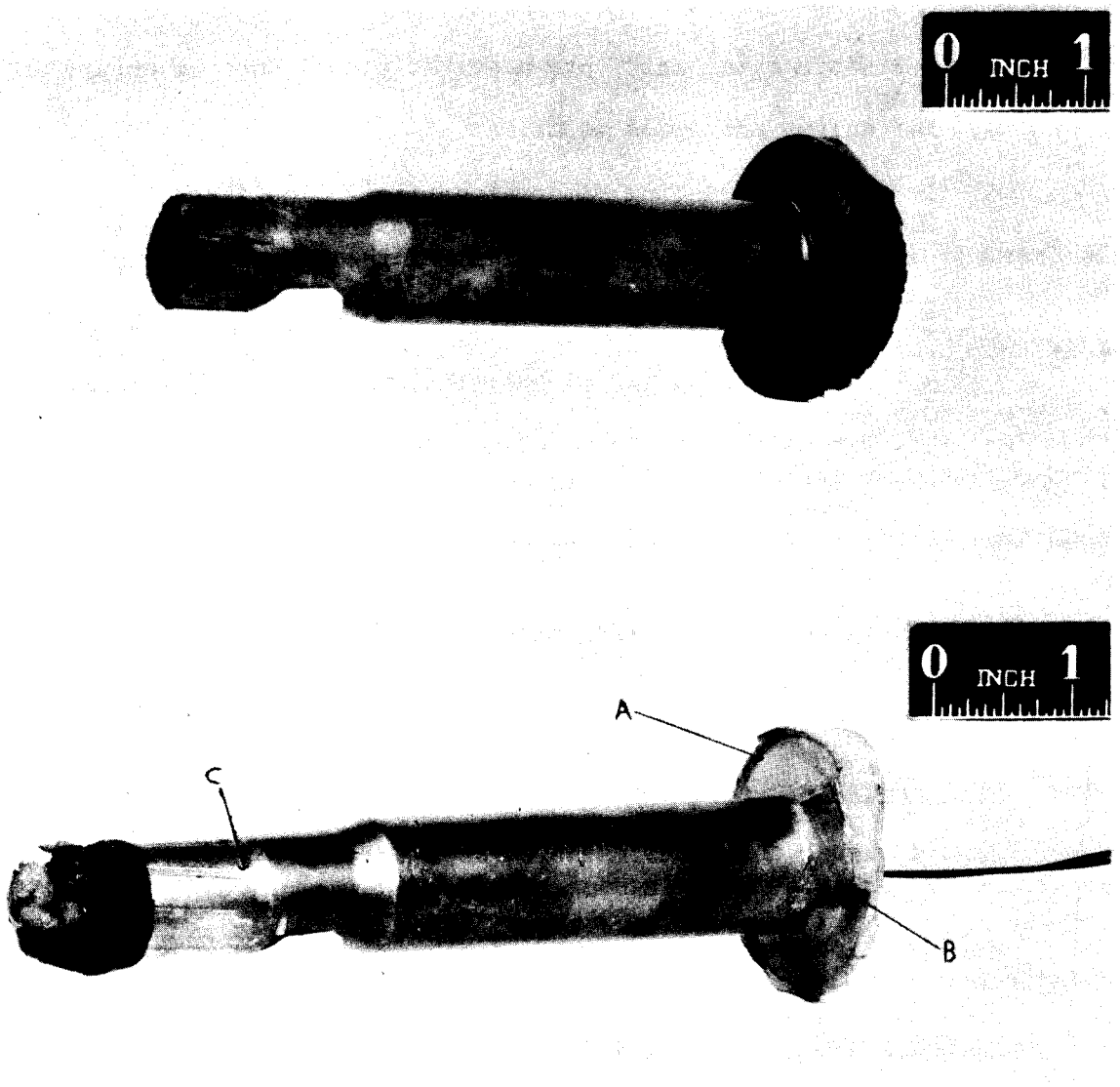


Figure 3. Casts of 20-mm Chambers.
(Top) - Sulfur Cast.
(Bottom) - Woods Metal Cast.

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APPENDIX A

DESCRIPTION AND/OR USES OF REQUIRED EQUIPMENT

FOR GUTTA-PERCHA IMPRESSIONS (See Figures A-1 and A-2)

a. Gutta-percha - a rubber-like resinous product. It becomes soft and pliable at temperatures ranging from 140° to 175°F and sets quickly at normal temperatures.
b. Powdered graphite - used to provide a smooth, uniform, black surface on the completed impression and to prevent it from sticking to bore.
c. Impression blocks - soft wood blocks used to mount the impression material. Blocks of various sizes are employed, dependent upon the nominal size of the cannon bore. For guns of calibers of 37-mm or less, a 6-inch long block is employed; for the larger weapons, the block should be at least 12 inches long. Additional dimensions for the blocks may be determined by reference to Figure A-2.
d. Wedges - employed to force the impression material into the contour of the area being surveyed. In general, the wedge is constructed on the same principle as the impression blocks.

FOR METAL CASTS

a. Sulfur casts - the sulfur cast material is made using a combination of eight (8) parts sulfur and one (1) part lampblack by weight. The melting point is 235°F.
b. Woods metal - the melting point varies from 150° to 250°F.

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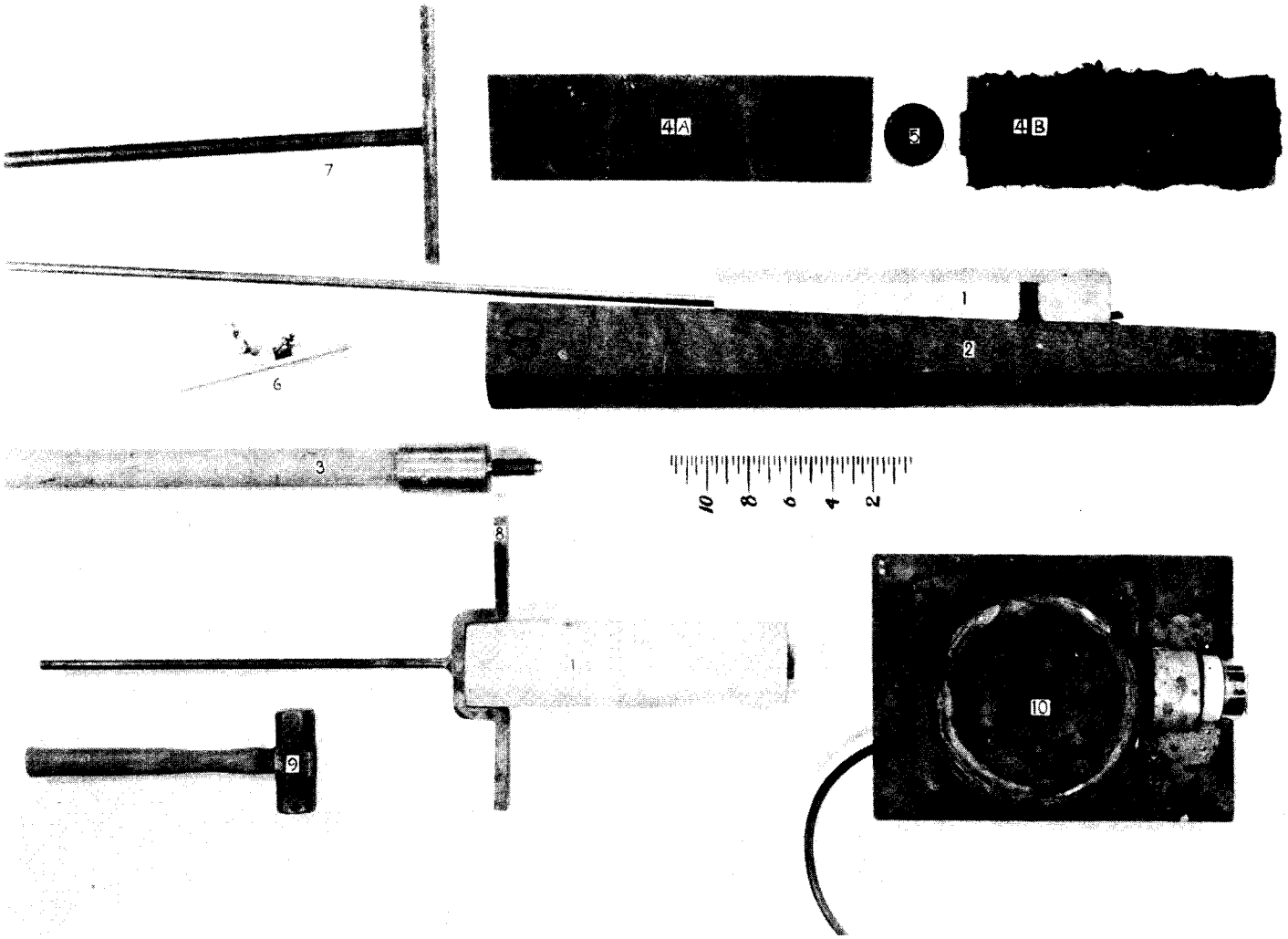
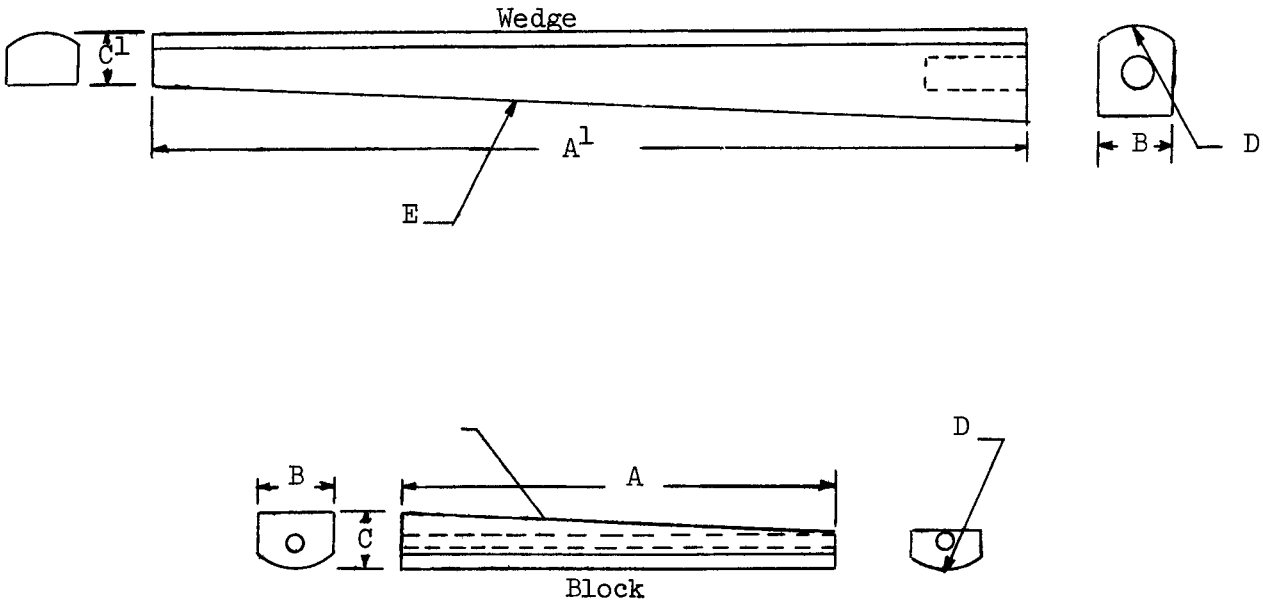


Figure A-1 Impression Making Device (Aberdeen Proving Ground). (1) Impression Block with Distance Rod, (2) Wedge, (3) Wood Rod for Wedge, (4a) Gutta-Percha (New), (4b) Gutta-Percha (Reclaimed), (5) Powdered Graphite, (6) Face Piece, (7) and (8) Special Face Plates for Impressions at End of Tubes, (9) Sledge, (10) Hot Plate. (Water Container not Shown)

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A - 6 inches for 37-mm or smaller bores
12 inches for bores larger than 37-mm
A^1 -Approximately 2 X A
B -2/3 diameter of bore
C -1/2 diameter of bore less 1/8 inch
C^1 -1/2 diameter of bore
D -Radius of bore less 1/8 inch
E -Taper of 1/2 inch per 12 inches

Figure A-2. Dimensions of Impression Block and Wedge Assembly

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